

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 6: Security, Privacy and Data Integrity

Subtopic 6.2: Data Integrity

Concept 6.2.1: Data Validation Methods

Total Questions: 10

Mark Schemes begin on Page 7

Question 2 | Source: 9618_w24_qp_11 (Q2.b.ii)

- 2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

PART(PartID, Description, Price, SupplierID)

```
CUSTOMER(CustomerID, FirstName, LastName, ContactNumber)
```

REPAIR(RepairNumber, StartDate, EndDate, CustomerID, Device)

REPAIR PART (PartID, RepairNumber, Quantity)

(ii) The table shows sample data for the table `REPAIR PART`.

PartID	RepairNumber	Quantity
ACD128SA	0022	3
PPOR543DWW	0022	1
TR453	0023	1
PPOR543DWW	0023	2
WED5	0024	5

Write a Structured Query Language (SQL) script to define the table `REPAIR PART`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

..... [5]

Question 3 | Source: 9618_s23_qp_12 (Q2.c.i)

- 2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.
- This database is created and managed using a Database Management System (DBMS).

(c) The database `Lessons` has the following tables:

```
HORSE(HorseID, Name, Height, Age, HorseLevel)

STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)

LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)
```

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(i) Describe **two** methods of validating the field `RiderLevel`.

1

.....

.....

2

.....

.....

[2]

Question 4 | Source: 9618_s23_qp_13 (Q4.d.i)

- 4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.
- (d) The data is validated and verified when it is entered into the database.

- (i) The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1

.....

2

.....

[2]

Question 5 | Source: 9618_s23_qp_13 (Q4.d.iii)

- 4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(d) The data is validated and verified when it is entered into the database.

- (iii) State why the car registration number might be incorrect even after it has been validated and verified.

.....

..... [1]

Question 6 | Source: 9618_w22_qp_12 (Q4.a)

- 4 (a) State the difference between **data verification** and **data validation**.

.....

.....

..... [1]

Question 7 | Source: 9618_w22_qp_12 (Q4.c)

- (c) One validation method is a presence check.

Describe **two other** validation methods that can be used to validate non-numeric data.

1

.....

2

.....

[2]

Question 8 | Source: 9618_w21_qp_11 (Q2.b.i)

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

.....

.....

Example

[2]

Question 9 | Source: 9618_s21_qp_11 (Q6.a)

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x
```

```
IF x < 0 OR x > 10 THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

Question 10 | Source: 9618_s21_qp_11 (Q6.b)

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(b)

```
INPUT x
```

```
IF x = "" THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

Question 11 | Source: 9618_s21_qp_11 (Q6.c)

- 6** Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(c)

```
INPUT x
```

```
IF NOT (x = "Red" OR x = "Yellow" OR x = "Blue") THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q7.e)

7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3
------	--	----------

Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q2.b.ii)

2(b)(ii)	1 mark for each bullet point • Create table REPAIR_PART with opening and closing brackets, all statements inside the brackets • PartID and RepairNumber as varchar (or equivalent) • Quantity as integer • At least 1 appropriate constraint • Dual primary key For example: <pre>CREATE TABLE REPAIR_PART(PartID VARCHAR(20) NOT NULL, RepairNumber VARCHAR(4) NOT NULL, Quantity INT NOT NULL, PRIMARY KEY (PartID, RepairNumber));</pre>	5
----------	--	----------

Mark Scheme for Question 3 | Source: 9618_s23_ms_12 (Q2.c.i)

2(c)(i)	1 mark each to max 2 • Presence check to make sure that the (rider level) is entered • Look-up / Existence check to make sure the rider level is only Beginner, Intermediate or Advanced • Length check to make sure the rider level entered is either 8 or 12 characters • Type check to make sure the rider level is alphanumeric	2
---------	---	----------

Mark Scheme for Question 4 | Source: 9618_s23_ms_13 (Q4.d.i)

4(d)(i)	1 mark each to max 2 - Length check: the registration number must be 6 characters long - Format check: the registration number must be in the format letter-digit-digit-digit-letter-letter - Type check: the registration number must be alphanumeric	2
---------	---	----------

Mark Scheme for Question 5 | Source: 9618_s23_ms_13 (Q4.d.iii)

4(d)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	1
-----------	--	----------

Mark Scheme for Question 6 | Source: 9618_w22_ms_12 (Q4.a)

4(a)	data verification is checking if input data is the same as the original whereas data validation is checking that the data is reasonable / sensible	1
------	--	----------

Mark Scheme for Question 7 | Source: 9618_w22_ms_12 (Q4.c)

4(c)	1 mark for each bullet point (max 2). For example: - to make sure data is in the required format // only expected characters allowed - to make sure the data is already present in the system - to make sure the data contains the correct number of characters - to ensure that non-numeric data is entered	2
------	--	----------

Mark Scheme for Question 8 | Source: 9618_w21_ms_11 (Q2.b.i)

2(b)(i)	1 mark per bullet point - validation checks that data is reasonable / sensible - example e.g. checking data is the right number / type of characters	2
---------	--	----------

Mark Scheme for Question 9 | Source: 9618_s21_ms_11 (Q6.a)

6(a)	Range (check)	1
------	---------------	----------

Mark Scheme for Question 10 | Source: 9618_s21_ms_11 (Q6.b)

6(b)	Presence (check)	1
------	------------------	----------

Mark Scheme for Question 11 | Source: 9618_s21_ms_11 (Q6.c)

6(c)	Existence (check)	1
------	-------------------	----------